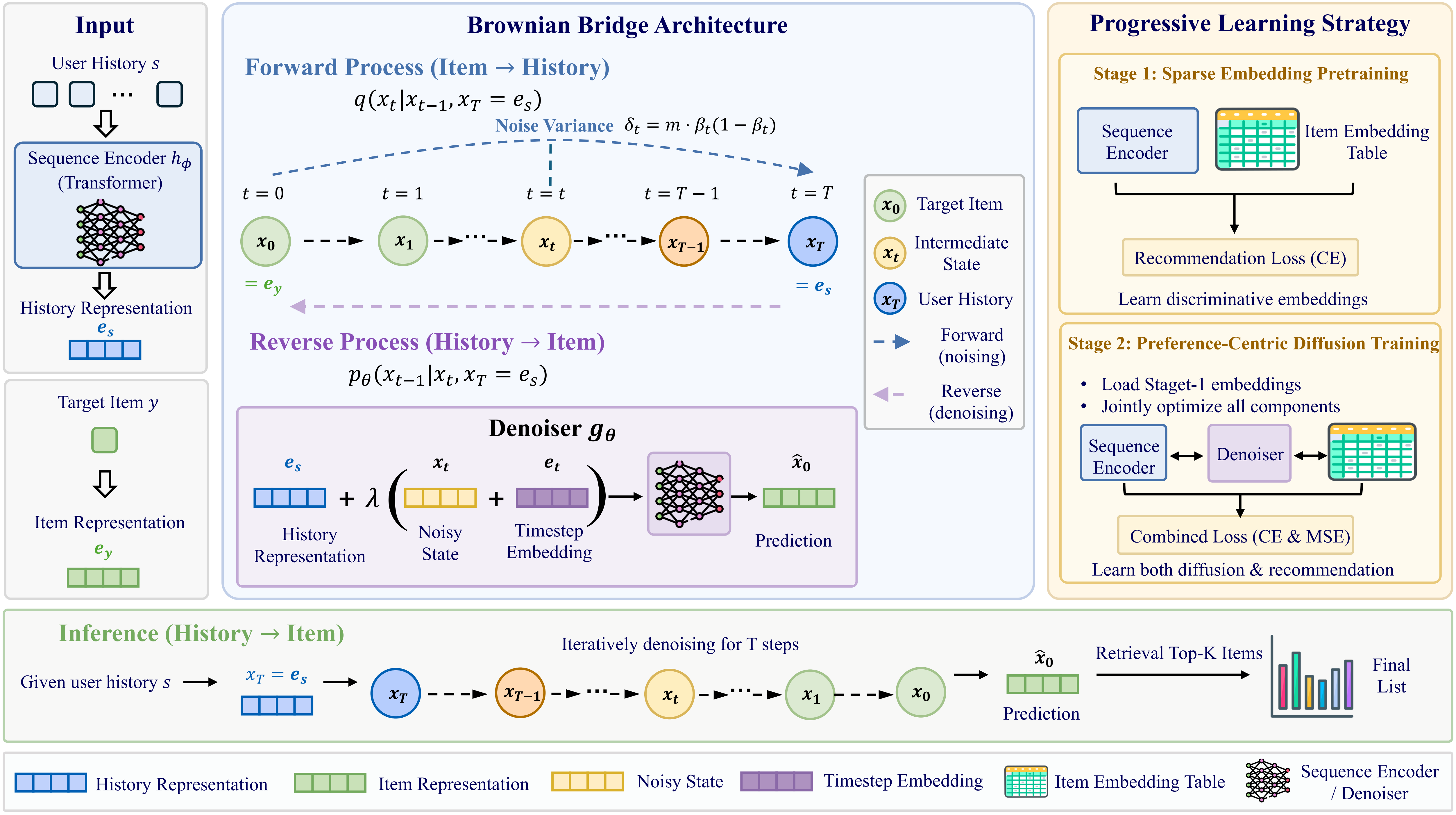

Input
User History s
Sequence Encoder h_ϕ (Transformer)
History Representation e_s
Target Item y
Item Representation e_y
Brownian Bridge Architecture
Forward Process (Item → History)
$q(x_t|x_{t-1}, x_T = e_s)$
Noise Variance $\delta_t = m \cdot \beta_t(1-\beta_t)$
$t=0$ $t=1$ $t=t$ $t=T-1$ $t=T$
x_0 x_1 x_t x_{T-1} x_T
$= e_y$ $= e_s$
Reverse Process (History → Item)
$p_\theta(x_{t-1}|x_t, x_T = e_s)$
Denoiser g_θ
$e_s + \lambda(x_t + e_t)$
History Representation
Noisy State
Timestep Embedding
$\hat{x}_0$
Prediction
Target Item
Intermediate State
User History
Forward (noising)
Reverse (denoising)
Progressive Learning Strategy
Stage 1: Sparse Embedding Pretraining
Sequence Encoder
Item Embedding Table
Recommendation Loss (CE)
Learn discriminative embeddings
Stage 2: Preference-Centric Diffusion Training
• Load Staget-1 embeddings
• Jointly optimize all components
Sequence Encoder
Denoiser
Combined Loss (CE & MSE)
Learn both diffusion & recommendation
Inference (History → Item)
Iteratively denoising for T steps
Given user history s
$x_T = e_s$
x_T x_{T-1} x_t x_1 x_0
$\hat{x}_0$
Prediction
Retrieval Top-K Items
Final List
History Representation
Item Representation
Noisy State
Timestep Embedding
Item Embedding Table
Sequence Encoder / Denoiser